

Lesson 5: Series and parallel

Electric Circuits · Year 10 · 50 minutes · AQA 6.2.2

Why this lesson is here

One broken lamp turns off a series chain. A parallel branch can keep the other lamp working.

Learning objective

Solve series and parallel circuit questions with clear reasons.

Core teaching point

Series: current equal, p.d.s add, resistances add. Parallel: p.d. equal across branches, branch currents add; total resistance falls when a branch is added.

Equipment and preparation

Cells, identical lamps and two resistors, meters; optional PhET lab; mini whiteboards.

Lesson sequence

Time	Phase	Teacher move and evidence
0-6	Retrieve	Measure/recall source and component p.d.s from Lab 2.
6-18	Model	Compare loop and branch diagrams; derive the rules.
18-31	Build and test	Measure currents and p.d.s in both arrangements.
31-43	Calculate	Tasks 2-4, using a rule before each calculation.
43-50	Check	Five-mark question and targeted misconception check.

Workbook task guidance

Task 1: Draw two lamps in series and two lamps in parallel. Mark junctions clearly.

Expected: Series: one loop. Parallel: two branches between the same junctions.

Task 2: Calculate the total resistance and current for 3.0 ohm and 9.0 ohm in series across 12 V.

Expected: $R_{\text{total}} = 12 \text{ ohm}$; $I = 12/12 = 1.0 \text{ A}$.

Task 3: Parallel branches carry 0.20 A and 0.35 A on a 6.0 V battery. State each branch p.d. and the supply current.

Expected: Each branch is 6.0 V; supply current = 0.55 A.

Task 4: Explain, without a parallel-resistance formula, why adding a branch lowers total resistance.

Expected: Extra path increases supply current at fixed p.d., so V/I for the whole circuit decreases.

Worked example

2.0 ohm and 4.0 ohm in series on 12 V: $R_{\text{total}} = 6.0 \text{ ohm}$; $I = 2.0 \text{ A}$; p.d.s = 4.0 V and 8.0 V.

AQA-style checkpoint

[5 marks] A 4.0 ohm and an 8.0 ohm resistor are in series on 12 V. Find total resistance, current and p.d. across each. State one parallel-circuit rule.

Mark guidance: $R_{\text{total}} = 12 \text{ ohm}$ (1); $I = 12/12 = 1.0 \text{ A}$ (1); $V_1 = 4.0 \text{ V}$ (1); $V_2 = 8.0 \text{ V}$ (1); e.g. same p.d. across branches (1).

Anticipated misconception

Current splits at a parallel junction; p.d. does not split between branches.

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.